

TECHNICAL DATASHEET

CPS-005 - SistemaVisao_Qualit
Cyber-Physical System Technical Datasheet

Prototype technical datasheet generated from the CPS Asset Administration Shell (AAS). It is not an official manufacturer datasheet.

1. CPS Identification

Attribute	Value	AAS source
CPS ID	CPS-005	assetInformation
Asset Name	SistemaVisao_Qualit	assetInformation
Asset Type	VisionInspectionSystem	assetInformation
Manufacturer	Custom/UFSC	assetInformation
Serial Number	SN-VIS-005	assetInformation
Global Asset ID	urn:asset:cps5:sistemavisao_qualit	assetInformation
AAS ID	urn:uuid:cps5-aas-001	AssetAdministrationShell
AAS idShort	CPS5_AAS	AssetAdministrationShell
Asset Kind	Instance	assetInformation

2. Technical Data

Attribute	Value	AAS source
ConveyorType	VisionInspectionSystem	TechnicalData
InspectionTechnology	CameraBasedVisualInspection	TechnicalData
SensorConfiguration	Luminosity, RPM, Torque, Temperature, CycleTime	TechnicalData
OperatingVoltage	24V DC	TechnicalData

3. Functional Capability

Attribute	Value	AAS source
Identifier	inspect_part_quality	Functions
Name	Inspect Part Quality	Functions.inspect_part_quality
Description	Inspeção visual da qualidade da peça com base em luminosidade, estabilidade operacional e desempenho do sistema de visão.	Functions.inspect_part_quality
Allowed statuses	active maintenance awaiting_replacement failure	Functions.inspect_part_quality

4. Communication Interfaces

Attribute	Value	AAS source
BrokerHost	broker.hivemq.com	AssetInterfacesDescription
BrokerPort	1883	AssetInterfacesDescription
BaseTopic	cps5	AssetInterfacesDescription
OEEInterface.Topic	cps5/oeo	OEEInterface
OEEInterface.Direction	Outbound	OEEInterface
ACSMOEEInterface.Topic	acsm/cps5/oeo	ACSMOEEInterface
ACSMOEEInterface.Direction	Outbound	ACSMOEEInterface
ACSMOEEInterface.Producer	CPS-5 / Node-RED	ACSMOEEInterface
ACSMOEEInterface.Consumer	ACSM	ACSMOEEInterface
ACSMLearningInterface.Topic	acsm/cps5/learning	ACSMLearningInterface
ACSMLearningInterface.Direction	Outbound	ACSMLearningInterface
ACSMLearningInterface.Producer	CPS-5 / Node-RED	ACSMLearningInterface
ACSMLearningInterface.Consumer	ACSM	ACSMLearningInterface
WebSocketInterfaces.MQTTWS	ws://broker.hivemq.com:8000/mqtt	WebSocketInterfaces
WebSocketInterfaces.MQTTWSS	wss://broker.hivemq.com:8884/mqtt	WebSocketInterfaces
RESTEndpoints.DescriptionEndpoint	http://localhost:1881/api/cps5/description	RESTEndpoints
RESTEndpoints.SummaryEndpoint	http://localhost:1881/api/cps5/summary	RESTEndpoints
RESTEndpoints.IndicatorsEndpoint	http://localhost:1881/api/cps5/indicators	RESTEndpoints
RESTEndpoints.HistoryEndpoint	http://localhost:1881/api/cps5/history	RESTEndpoints
RESTEndpoints.HealthEndpoint	http://localhost:1881/api/cps5/health	RESTEndpoints

5. Lifecycle Integration

Lifecycle phases are distinct from operational conditions. Plug, Play-Evolution, and Unplug are lifecycle phases; running, stopped, and maintenance are operational conditions and are not treated as lifecycle phases in this document.

Attribute	Value	AAS source
SupportedPhases	plug, play, monitor, unplug	LifecycleIntegration
CurrentPhase	play	LifecycleIntegration

6. ACSM Integration

Attribute	Value	AAS source
DashboardURL	http://localhost:1881/ui	ACSMIntegration
IntegrationMode	MQTT + REST	ACSMIntegration
LifecycleOrchestration	ACSM-driven	ACSMIntegration
AnalyticsIntegrationMode	MQTT + AAS	ACSMIntegration
OEEPublishTopic	acsm/cps5/oe	ACSMIntegration
LearningPublishTopic	acsm/cps5/learning	ACSMIntegration
OEESubmodelRef	urn:uuid:submodel:cps5:oeemonitoring	ACSMIntegration
LearningSubmodelRef	urn:uuid:submodel:cps5:learning	ACSMIntegration

7. Digital Representation

Attribute	Value	AAS source
Representation	Asset Administration Shell	AssetAdministrationShell
Global Asset ID	urn:asset:cps5:sistemavisao_qualit	assetInformation
AAS instance ID	urn:uuid:cps5-aas-001	AssetAdministrationShell
Document endpoint	/api/cps5/datasheet/pdf	Documents.DatasheetPDF

8. Document Information

Attribute	Value	AAS source
Document type	Prototype Technical Datasheet	Generated document
Source	CPS-005 Asset Administration Shell	Source definition
Generated for	PPU Prototype	Requested document scope
Date generated	2026-08-30	Generation date

Excluded runtime data

Current temperature, RPM, torque, luminosity, counters, cycle time, OEE, availability, performance, quality, health, Learning, Reasoning, Prediction, and Recommendation are runtime data or intelligence outputs. They are deliberately excluded from this static technical specification.